

RUMOI, Japan

WMO: 474060

Lat: 43.9461N

Lon: 141.6319E

Elev: 83

StdP: 14.65

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.8	13.2	3.8	6.7	12.6	6.7	7.8	15.7	34.2	26.7	31.2	25.3	12.4	110	0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.9	80.6	72.5	77.8	71.0	75.7	70.0	74.0	78.6	72.6	76.5	71.2	74.9	9.0	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.5	120.9	77.0	71.2	115.6	75.7	69.8	110.0	74.3	37.7	78.7	36.5	76.3	35.2	75.0	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.1	26.8	23.9	DB	2.9	86.7	4.2	3.0	-0.2	88.9	-2.6	90.7	-5.0	92.4	-8.1	94.6	(4)
(5)			WB	2.2	76.3	4.2	2.1	-0.8	77.9	-3.2	79.1	-5.6	80.3	-8.6	81.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.4	24.1	24.9	31.5	41.7	52.2	59.8	67.5	70.1	63.1	51.8	39.8	29.2		
	DBStd	16.92	5.30	5.84	5.58	5.89	6.25	5.23	4.61	4.41	5.50	5.49	6.53	5.41		
	HDD50	3389	803	702	573	260	47	2	0	0	0	44	312	646		
	HDD65	7106	1268	1122	1038	699	400	168	25	9	100	410	756	1111		
	CDD50	2085	0	0	0	11	114	297	543	623	392	99	6	0		
	CDD65	327	0	0	0	0	2	14	103	166	42	0	0	0		
	CDH74	902	0	0	0	0	10	28	223	526	115	0	0	0		
	CDH80	106	0	0	0	0	1	2	30	59	14	0	0	0		
	WSAvg	11.5	14.1	13.3	12.4	10.9	9.7	7.9	7.8	8.7	10.2	12.8	14.5	15.3		
	PrecAvg	46.0	4.0	2.9	2.2	1.9	2.5	2.3	4.5	4.8	5.9	4.8	5.4	4.8		
(15) Precipitation	PrecMax	57.7	5.7	5.3	3.7	3.9	4.4	4.3	8.9	8.7	14.3	8.7	8.9	7.6		
	PrecMin	35.5	2.3	1.4	0.6	0.5	0.7	0.9	1.0	1.6	2.4	1.8	3.3	2.5		
	PrecStd	5.7	1.0	1.0	0.8	0.9	0.9	0.7	2.2	1.9	3.0	1.7	1.6	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.7	42.8	50.3	66.3	74.7	77.3	83.8	85.1	81.6	69.3	60.7	47.8		
		MCWB	36.2	37.8	42.1	52.0	59.0	65.4	73.8	75.5	72.5	60.5	53.6	44.2		
	2%	DB	36.3	38.7	45.3	59.2	68.9	73.1	79.3	81.4	77.0	66.0	56.4	42.7		
		MCWB	33.1	35.0	39.6	48.9	57.2	64.1	71.7	73.7	69.9	58.9	51.0	38.9		
	5%	DB	34.0	36.3	42.7	54.6	64.9	69.6	76.1	78.8	74.2	63.7	53.5	39.8		
		MCWB	31.2	32.8	38.2	46.5	55.7	62.9	70.4	72.2	68.0	56.9	48.4	36.1		
	10%	DB	32.0	33.9	40.2	50.9	61.3	66.9	73.9	76.8	72.0	61.3	50.6	37.3		
		MCWB	29.2	30.8	36.2	44.9	54.3	61.5	69.5	71.2	66.2	55.2	45.9	33.6		
	0.4%	WB	36.5	38.5	43.5	53.9	61.6	68.4	75.5	76.5	73.7	62.9	54.8	44.1		
		MCDB	39.1	41.7	48.4	62.4	69.9	74.0	82.0	82.4	79.3	66.9	59.5	47.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	33.4	35.4	40.6	50.2	58.9	65.5	73.1	74.7	71.5	60.2	51.8	39.4		
		MCDB	35.5	38.4	44.0	57.0	65.7	70.4	76.8	79.6	75.8	64.5	55.7	42.2		
	5%	WB	31.6	33.2	38.6	47.8	57.0	63.8	71.6	73.5	69.2	58.0	49.0	36.5		
		MCDB	33.8	36.0	42.2	53.3	63.2	68.0	75.2	77.3	72.9	62.5	53.1	39.4		
	10%	WB	29.4	30.9	36.6	45.5	55.1	62.3	70.0	72.2	67.0	55.8	46.1	33.9		
		MCDB	31.7	33.6	40.0	50.3	60.5	66.1	73.1	75.9	71.2	60.3	50.2	36.7		
	5% DB	MDBR	10.2	11.2	10.9	12.8	13.3	11.0	9.9	10.9	13.7	13.8	10.7	9.2		
		MCDBR	11.3	12.8	14.5	19.8	19.4	15.3	13.1	12.9	14.8	16.4	16.0	12.1		
		MCWBR	10.1	11.1	10.7	11.6	10.2	8.2	7.0	7.1	8.7	10.7	12.6	10.5		
		MCWBR	11.4	12.5	13.7	17.9	17.1	12.9	11.4	10.8	12.7	14.7	15.2	12.2		
(35) Mean Daily Temperature Range	5% WB	MCWBR	10.5	11.4	10.8	11.4	9.6	7.5	6.3	6.3	8.2	10.5	12.7	11.0		
	Clear-Sky Solar Irradiance	taub	0.352	0.405	0.465	0.525	0.538	0.522	0.500	0.465	0.414	0.414	0.398	0.371		
		taud	2.021	1.885	1.805	1.792	1.858	1.984	2.113	2.231	2.318	2.156	2.092	1.998		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	228	235	238	236	238	243	247	252	256	234	211	205		
		Edh at Noon	35	48	60	65	63	56	49	42	36	36	33	33		
(44) All-Sky Solar Radiation	RadStd		342	593	988	1441	1647	1633	1521	1436	1184	781	405	270		
			33	69	101	141	137	173	180	102	102	56	52	26		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page